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Random coefficient differential equation models for bacterial growth

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ABSTRACT

In the mathematical modeling of population growth, and in particular of bacterial growth, parameters are either measured directly or determined by curve fitting. These parameters have large variability that depends on the experimental method and its inherent error, on differences in the actual population sample size used, as well as other factors that are difficult to account for. In this work the parameters that appear in the Monod kinetics growth model are considered random variables with specified distributions. A stochastic spectral representation of the parameters is used, together with the polynomial chaos method, to obtain a system of differential equations, which is integrated numerically to obtain the evolution of the mean and higher-order moments with respect to time.

1. Introduction

In order to get a better understanding of the processes involved, to extend results and to make predictions, mathematical models are indispensable in many areas of science and engineering. For bacterial growth the models usually take the form of differential equations or systems thereof. The coefficients of these equations have traditionally been deterministic, that is they have been assumed to be known and have no variation. Differential equations with deterministic coefficients have been extensively studied, both from the analytical and numerical viewpoints. However, in many situations, equations with random coefficients are better suited in describing the real behavior of quantities of interest than their counterpart with deterministic coefficients. Randomness in the coefficients may arise because of errors in the observed or measured data, variability in experiment and empirical conditions, uncertainties (variables that cannot be measured, missing data, etc) or plainly because of lack of knowledge. Differential equations where some or all the coefficients are considered random variables or that incorporate stochastic effects (usually in the form of white noise) have been increasingly used in the last few decades to deal with errors and uncertainty and represent a growing field of great scientific interest, see [1,2].

Monte Carlo methods [3,4] have been used for a long time to perform simulations when random effects were involved. They are simple to implement and understand but require many realizations due to their slow convergence rate and hence tend to be expensive. Other methods that have been developed and used are the moment method [5,6] and the polynomial chaos method, see [7] and the references therein. In this paper we will use the latter approach to study the time evolution of a system governed by the Monod kinetic equations [8] under a variety of random inputs.

The growth of microorganisms such as bacteria, fungii and algae is of great interest in biology and medicine. Field observations and laboratory experiments are expensive and very difficult to perform; in the field it is almost impossible to keep the external factors constant and uniform. In the laboratory there is much more control; however, the conditions may still vary in time and also be different from actual situations in the real environments of interest. Moreover, errors in measuring the population sizes occur frequently. Even when measurements are done with the utmost care, the measured

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values will differ between experiment batches; most times the variability is quite dramatic. This is due to the large sizes and/or variability of the populations, inaccuracies in the methods used to assess them, error (human or otherwise), as well as other unknown factors. In medicine all these difficulties are increased by orders of magnitude. Thus, the importance of mathematical models in the life sciences cannot be overstated. For a more realistic approach to the problem, the parameters involved in these models, such as the growth rate for example, need to be treated as random quantities. Two population models still in use, albeit relatively old, are the exponential growth (or Malthusian growth) and the logistic growth. Random coefficient differential equation models for such growth that have been solved numerically using polynomial chaos were recently explored in [9]. In this paper we consider a more realistic model for microbial growth, first proposed by Monod [8] and based on the assumption that the growth rate depends on the amount of available nutrients. The Monod model is widely used; specifically, it is used for biofilm growth. Biofilm-forming bacteria are bound together by an extracellular polymer substance (*EPS*) [10] and become firmly attached to the solid surfaces. They can be used to form bio-barriers in porous media, that in turn control the spread of contaminants in aquifers [11]. Models for biofilm growth and/or convection are therefore of large interest.

Here, as a first step, the bacteria and nutrients are considered immobile, but the parameters of the model are allowed to be independent random variables with various distributions. Equations for the time evolution of their polynomial chaos expansion coefficients are obtained and solved numerically. From these coefficients, means, variances and even higher-order statistics can be obtained. The paper is structured as follows. In Section 2, the Monod mathematical model for bacterial growth is briefly described. The polynomial chaos approach is presented in Section 3; a discussion of the types of chaos considered herein versus the different types of random inputs is undertaken in Section 4. Section 5 is devoted to numerical results. Finally, our conclusions in Section 6 end the paper.

2. Monod kinetic models

If we let $y(t)$ be the population of a given species at time t, then the simplest model for the growth or decay of the population is that the rate of change is proportional to the size of the population. This is the model proposed by Thomas Malthus in 1798 [12,13]. The differential equation governing $y(t)$ for this model is

$$\frac{\mathrm{d}y}{\mathrm{d}t} = ry(t), \tag{1}$$

where r is the growth rate. Given an initial value of the population $y(t=0) = y_0$, the solution is

$$y(t) = y_0 \exp(rt). \tag{2}$$

Obviously, as the population keeps growing inside the test tubes, there starts to be a competition for the limited resources. In 1838, Verhulst [14] proposed that the growth rate should decrease with the size of the population. This in turn leads to the logistic equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = r\left(1 - \frac{y}{K}\right)y. \tag{3}$$

Here r still denotes the growth rate and K is the equilibrium value. The solution, subject to initial value $y(0) = y_0$, is

$$y(t) = \frac{y_0 K}{y_0 + (K - y_0)\exp(-rt)}. \tag{4}$$

This model fits experimental data surprisingly well in many cases, although it only considers that the competition for resources introduces a quadratic term limiting the growth. It does not offer a physical, chemical or biological explanation for this quadratic term. On the contrary, Monod [8] proposed a model in which the rate growth depends on the amount of necessary nutrients. If the nutrients are present in abundance, the model resembles Malthusian growth; as the amount of nutrients decreases, so does the growth rate. The Monod model is more realistic and consequently more complicated. There are at least two coupled differential equations, one for the microorganism population and the other for the amount of nutrients.

Here we consider a Monod type microbial growth for microorganisms in test tubes, where there is essentially no convection or diffusion of either microbes or nutrients. Both microbes and nutrients are supposed to reside in a liquid phase, usually water. In the case when there is only one necessary nutrient (or equivalently only one nutrient limits the growth), Monod's model is given by:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(c_M) &= r_M(c_M, c_N)(\text{microbes}), \\ \frac{\mathrm{d}}{\mathrm{d}t}(c_N) &= r_N(c_M, c_N)(\text{nutrients}). \end{aligned} \tag{5}$$

Here c_i, $i = M, N$, represents the mass concentration of species i per unit volume of the liquid phase; r_i represents the total rate at which species i is produced via reactions and sources. The molecular species present are the microbes, labeled M, and the soluble nutrients, labeled N. Furthermore, the microbial death rate is assumed proportional to the size of the population. The rate of microbial growth is given by the Monod kinetics reactions:

$$\mu(c_N) = \frac{\mu_{\mathrm{m}} c_N}{K_s + c_N}, \tag{6}$$

where μ_{m} is the maximum specific growth rate and K_s is that value of the concentration of nutrients c_N where the specific growth rate $\mu(c_N)$ has half its maximum value [15].

Invoking all simplifying assumptions, Eq. (5) lead to the following system of differential equations:

$$\begin{aligned} \frac{\mathrm{d}c_M}{\mathrm{d}t} &= \frac{\mu_{\mathrm{m}} c_N}{K_s + c_N} c_M - k_r c_M, \\ \frac{\mathrm{d}c_N}{\mathrm{d}t} &= \frac{\mu_y c_N}{K_s + c_N} c_M, \end{aligned} \tag{7}$$

where k_r is the first-order endogenous decay rate and μ_y is a negative coefficient that incorporates the effect of the yield rate. Eq. (7) represents a coupled system of nonlinear ordinary differential equations with parameters μ_{m}, K_s, k_r and μ_y; these equations need to be solved numerically even in the case of deterministic parameters. If the parameters are considered random variables the situation obviously becomes more complicated. While in principle all parameters as well as the initial conditions for c_M and c_N may be considered random, in order to make the model more tractable and also to simplify the presentation we will consider a three-dimensional random parameter space. Since initial values of the concentrations of microbes and nutrients can usually be measured much more accurately than their values later on, we will consider these initial values to be deterministic. Another reason for considering these values deterministic is that, when performing the necessary curve fitting that establishes the mean values of the parameters based on measurements of the amounts of microbes and nutrients, the initial values are usually not part of the fitting process. Thus, the parameters that are considered random in the following are μ_{m}, μ_y and K_s. An additional simplification is achieved by the assumption that these parameters are independent. While this assumption is not needed for a numerical solution, it will greatly improve the readability and treatment of the governing discrete equations. Furthermore, the endogenous decay rate is set to $k_r = 0$. The motivation for this simplification comes from the fact that the amount of nutrients in a test tube is usually consumed in a much shorter time than is necessary for the natural death of bacteria to contribute any significant changes in the population. The next section presents our approach to the numerical solution of the Monod model with the random coefficients thus identified using polynomial chaos.

3. Random coefficients and polynomial chaos

Given the stated assumptions, the Monod kinetic model that will be explored in the following can be written in the form

$$\begin{aligned} \frac{\mathrm{d}c_M}{\mathrm{d}t} &= \frac{\mu_{\mathrm{m}} c_N}{K_s + c_N} c_M, \\ \frac{\mathrm{d}c_N}{\mathrm{d}t} &= \frac{\mu_y c_N}{K_s + c_N} c_M \end{aligned} \tag{8}$$

with the maximum specific growth rate μ_{m}, the half-growth concentration rate K_s and the quantity μ_y allowed to be random variables. Thus, the values of these parameters are supposed to depend on the outcome ω of an experiment, $\mu_{\mathrm{m}} = \mu_{\mathrm{m}}(\omega)$, $\mu_y = \mu_y(\omega)$ and $K_s = K_s(\omega)$, where ω takes values in the set of all outcomes Ω. The latter is assumed to be properly equipped with a σ-algebra $\mathcal{F}$ and a probability measure P such that the triple $(\Omega, \mathcal{F}, P)$ forms a probability space [16]. The concentrations $c_M(t;\omega)$ and $c_N(t;\omega)$ then become stochastic processes. To develop a general methodology for the numerical solution of the evolution Eq. (8) as well as for estimating the various moments of the solution, we follow a Generalized Polynomial Chaos (GPC) approach [17,18,7]. In this context, a random quantity $\chi(\omega)$ is projected on the space of polynomial chaos

$$\chi(\omega) = \chi_0 \Gamma_0 + \sum_{i_1=1}^{\infty} \chi_{i_1} \Gamma_1(\xi_{i_1}(\omega)) + \sum_{i_1=1}^{\infty} \sum_{i_2=1}^{i_1} \chi_{i_1 i_2} \Gamma_2(\xi_{i_1}(\omega), \xi_{i_2}(\omega)) + \cdots \tag{9}$$

where the Γ_i are successive polynomial chaos, of increasing degree, in their arguments [19,20,17]. This expansion has been shown to converge, for the particular case of Hermite expansion, for second-order random processes [21]. Other orthogonal polynomials and their relationship with various distributions of random variables are discussed in the paper by Xiu and Karniadakis [7].

The polynomial chaos can be arranged in a sequence $\Phi_i(\boldsymbol{\xi}(\omega))$, such that the expansion of the random variables and stochastic processes appearing in Eq. (8) takes the form, i.e.:

$$c_M(t;\omega) = \sum_{i=0}^{\infty} c_{Mi}(t) \Phi_i(\boldsymbol{\xi}(\omega)); \qquad K_s(\omega) = \sum_{j=0}^{\infty} K_{sj} \Phi_j(\boldsymbol{\xi}(\omega)) \tag{10}$$

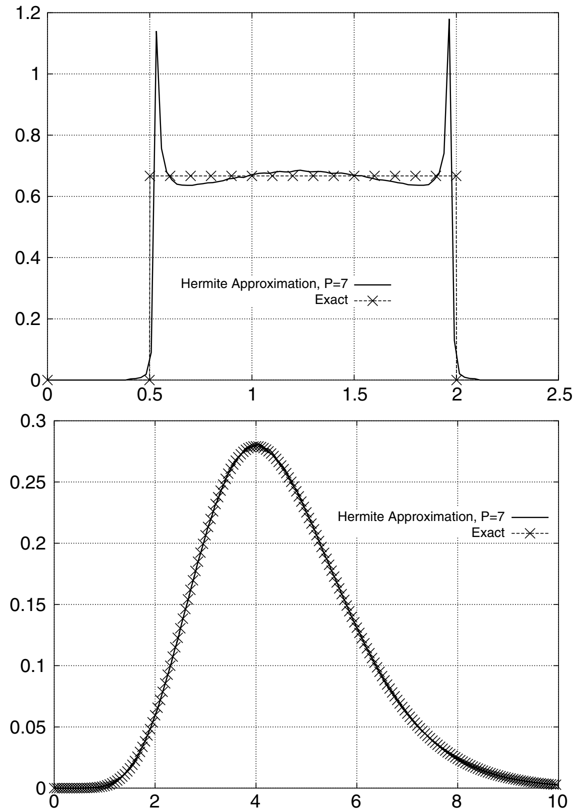

Fig. 1. Non-Gaussian input representation by Hermite chaos. Top figure: uniformly distributed $U(0.5, 2)$ random variable. Bottom figure: Gamma(9, 0.5) random variable.

where the Φ_i are properly chosen polynomial basis functions of the random variable vector $\boldsymbol{\xi}$. The number of variables in $\boldsymbol{\xi}$ represents the dimension of the chaos. We will perform two types of expansions, in terms of Jacobi polynomials and Hermite polynomials (the usual Wiener chaos expansion). They will be applied to the two different types of probability distribution functions that can occur for the parameters, with or without compact support, respectively. In the first case $\boldsymbol{\xi}(\omega)$ is a vector of random variables with beta distribution (the uniformly distributed random variable being a special case corresponding to Legendre polynomials as special case of the Jacobi polynomials); in the second case it is a vector of standard Gaussian random variables. A Galerkin projection using the orthogonality of the basis functions $\langle \Phi_i, \Phi_j\rangle = \delta_{ij}\langle \Phi_i, \Phi_i\rangle$, together with truncation of the polynomial chaos series to a finite number of terms will then lead to a system of ordinary differential equations governing the time evolution of the chaos coefficients of the solutions c_{Mi}, c_{Ni} of the system of differential Eq. (8).

A proper description of the random parameters in terms of the independent chaos variables in $\boldsymbol{\xi}$ must take into account the correlation between these parameters. Since we assume the three parameters to be independent random variables, each of them will be expanded as a functional of only one separate variable in $\boldsymbol{\xi}$; their random space is one-dimensional. More precisely, writing our expansions for these parameters only up to order one for readability, we will have

$$\begin{aligned}
\mu_{\mathrm{m}} &= \bar{\mu}_{\mathrm{m}} + \mu_{\mathrm{m}1}\Gamma_1(\xi_1) + \mu_{\mathrm{m}2}\Gamma_1(\xi_2) + \mu_{\mathrm{m}3}\Gamma_1(\xi_3)\\
\mu_y &= \bar{\mu}_y + \mu_{y1}\Gamma_1(\xi_1) + \mu_{y2}\Gamma_1(\xi_2) + \mu_{y3}\Gamma_1(\xi_3)\\
K_s &= \bar{K}_s + K_{s1}\Gamma_1(\xi_1) + K_{s2}\Gamma_1(\xi_2) + K_{s3}\Gamma_1(\xi_3)
\end{aligned} \tag{11}$$

where $\mu_{\mathrm{m}2} = \mu_{\mathrm{m}3} = 0$, $\mu_{y1} = \mu_{y3} = 0$ and $K_{s1} = K_{s2} = 0$ such that the three random parameters are uncorrelated. On the other hand, the expansions for the output stochastic processes c_M and c_N become three-dimensional chaos expansions. For c_M for example, the expansion will take the form

$$c_M(t;\omega) = \bar{c}_M(t) + c_{M1}(t)\Gamma_1(\xi_1) + c_{M2}(t)\Gamma_1(\xi_2) + c_{M3}(t)\Gamma_1(\xi_3). \tag{12}$$

For all random quantities, the first coefficient in the expansion represents the mean and will be denoted either by subscript zero or by an overbar, i.e. $\bar{c}_M(t) = c_{M0}(t)$.

We are now ready to develop the discrete equations used in the numerical study. The first equation in (8) is first written in the form:

$$(K_s + c_N)\,\frac{\mathrm{d}c_M}{\mathrm{d}t} = \mu_{\mathrm{m}} c_N c_M.$$

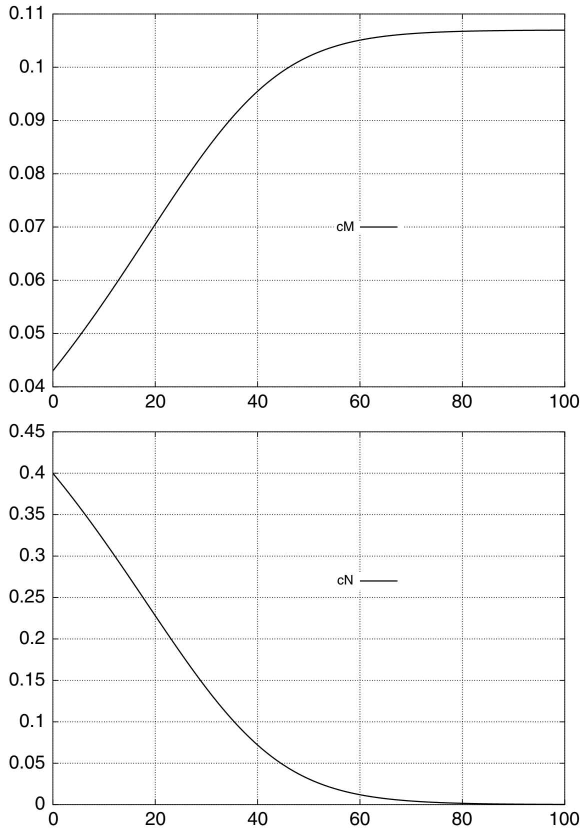

Fig. 2. Deterministic results for the concentrations; Top figure: c_M. Bottom figure: c_N.

Introducing the polynomial chaos expansions leads to

$$\left(\sum_{k=0}^{P} K_{sk}\Phi_k + \sum_{j=0}^{P} c_{Nj}\Phi_j\right)\sum_{i=0}^{P}\frac{\mathrm{d}c_{Mi}}{\mathrm{d}t}\Phi_i = \sum_{l=0}^{P}\sum_{j=0}^{P}\sum_{i=0}^{P} c_{Mi}c_{Nj}\mu_{\mathrm{m}l}\Phi_i\Phi_j\Phi_l \tag{13}$$

where an appropriate ordering of the basis functions Φ_i is used and their dependence on the random variables $\boldsymbol{\xi}$ results from the ordering and is not indicated explicitly. A Galerkin projection on the respective Hilbert space of random variables is obtained by demanding that the residual be orthogonal on the subspace spanned by the basis functions. In particular, taking the inner product of the above equation with basis function Φ_L results in

$$\sum_{i=0}^{P}\sum_{j=0}^{P}\left(K_{sj} + c_{Nj}\right)\langle\Phi_i\Phi_j, \Phi_L\rangle\frac{\mathrm{d}c_{Mi}}{\mathrm{d}t} = \sum_{i=0}^{P}\sum_{j=0}^{P}\sum_{l=0}^{P} c_{Mi}c_{Nj}\mu_{\mathrm{m}l}\langle\Phi_i\Phi_j\Phi_l, \Phi_L\rangle. \tag{14}$$

This is seen to be a matrix equation for the vector of unknown variables

$$\vec{c}_M = \begin{bmatrix} c_{M0} \\ c_{M1} \\ \vdots \\ c_{MP} \end{bmatrix} \tag{15}$$

of the form $A\mathrm{d}(\vec{c}_M)/\mathrm{d}t = \vec{b}_M$, where the matrix A and the vector $\vec{b}_M$ are themselves time-dependent. The entry on line L, column i of the matrix A is given by

$$A[L, i] = \sum_{j=0}^{P}\left(K_{sj} + c_{Nj}\right)\langle\Phi_i\Phi_j, \Phi_L\rangle.$$

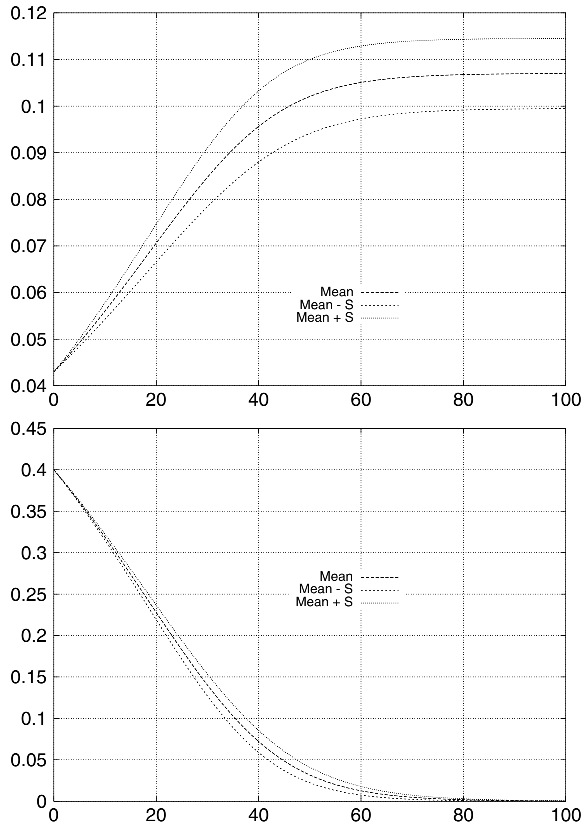

Fig. 3. Results for the mean concentrations and their variance for Case 1. Top figure: c_M. Bottom figure: c_N.

A similar equation $A\mathrm{d}(\vec{c}_N)/\mathrm{d}t = \vec{b}_N$, with the same matrix A, is obtained for the c_N part of the solution, with μ_y on the right-hand side in place of μ_m. The two equations thus lead to a system of the form

$$\begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t} \begin{bmatrix} \vec{c}_M \\ \vec{c}_N \end{bmatrix} = \begin{bmatrix} \vec{b}_M \\ \vec{b}_N \end{bmatrix}. \tag{16}$$

Upon use of an explicit Runge–Kutta method and linearization of the coefficient matrix and the right-hand side values using lagged values of the coefficients from the previous stage, the matrix can be easily inverted to obtain updated coefficient values.

4. Random input representation

In most situations a subset of the random input coefficients of the model must satisfy constraints inherent to the model itself. For example, in the case of the Monod model that we are using (8), the parameter K_s must be necessarily positive; thus assuming a normal distribution for K_s is not appropriate. It could be argued that μ_m should also be positive for growth to occur; however, this parameter may be allowed to have a normal distribution with a positive mean. In this case the implicit assumption is that the growth conditions are varying to such an extent that population decay instead of growth may sometimes occur, possibly because of other factors not accounted for in the model. The amount of nutrients is expected to decrease continuously, therefore the parameter μ_y should be negative.

If all the input parameters are random variables with the same type of distribution, the polynomial chaos basis is easy to choose. For example, if all the three parameters in the model were assumed to have normal distributions, a Hermite chaos expansion would be a natural choice. On the other hand, if all the three parameters were assumed to have uniform distributions, a Legendre chaos expansion would be preferred. Note however that even in this case the output distribution is not necessarily best described by these choices of basis functions. Even more importantly, the three parameters will likely have different distributions. Consider for example the perfectly valid case where K_s is assumed to have an uniform distribution, $-\mu_y$ a Gamma distribution, while μ_m has a normal distribution, and suppose one chooses to use the Hermite chaos expansion. Then obviously a first step in the solution process is to represent K_s and $-\mu_y$ using a Hermite chaos expansion. Here we describe our methodology for doing so, which is based on mappings from other distributions to the

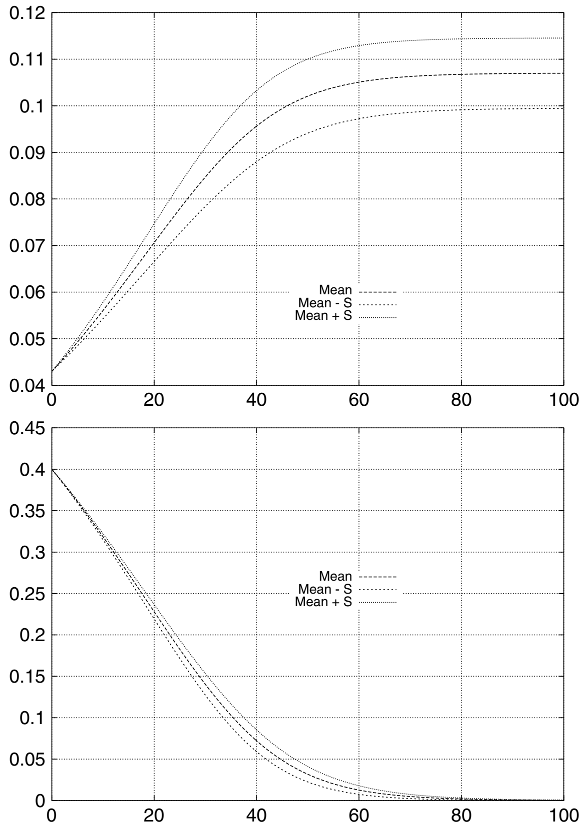

Fig. 4. Results for the mean concentrations and their variance for Case 2. Top figure: c_M. Bottom figure: c_N.

normal. Specifically, if ξ is a standard normal random variable then:

$$u(\xi) = a + (b-a)\frac{1+\mathrm{erf}(\xi/\sqrt{2})}{2}$$

has a uniform $U(a, b)$ distribution, and

$$\gamma(\xi) = ab\left(\xi\sqrt{\frac{1}{9a}} + 1 - \frac{1}{9a}\right)^3$$

has a Gamma(a, b) distribution. The Hermite coefficients of a random variable η for which a mapping from the standard normal is known can be readily computed by performing the inner product with Gaussian measure,

$$\eta_i = \frac{1}{\langle \Phi_i^2 \rangle}\langle\, \eta(\xi), \Phi_i(\xi)\, \rangle. \tag{17}$$

For the case of Legendre chaos expansions, the variables ξ are uniformly distributed random variables. The above method in this case can be applied easily if the cumulative distribution function of the variable $\eta(\xi)$ can be inverted (i.e. one uses the inverse transform method [16]) to map the random variable η on the probability space of uniformly distributed random variables. Given the cumulative distribution function $F_\eta(x)$ and ξ uniformly distributed in $(0, 1)$, the random variable $F_\eta^{-1}(\xi)$ has the same distribution as η. The inner products of η with the basis functions can then be computed as:

$$\eta_i = \frac{1}{\langle \Phi_i^2 \rangle}\langle \eta, \Phi_i \rangle = \frac{1}{\langle \Phi_i^2 \rangle}\int_0^1 F_\eta^{-1}(\xi)\Phi_i(\xi)\mathrm{d}\xi. \tag{18}$$

Fig. 1 shows the results obtained by representing a $U(0.5, 2)$ and a Gamma(a, b) random variable with probability density function given by

$$F_\eta(x) = \frac{x^{a-1}\exp(-x/b)}{\Gamma(a)b^a}$$

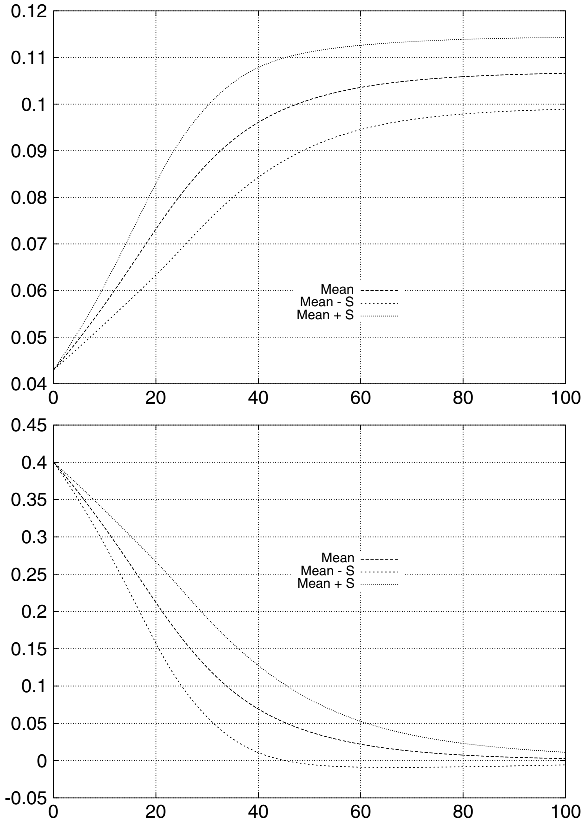

Fig. 5. Results for the mean concentrations and their variance for Case 3. Top figure: c_M. Bottom figure: c_N.

with $a = 9$, $b = 0.5$, respectively, by the Hermite seventh-order chaos. The approximate curve has been obtained numerically by plotting the histogram of one million Monte Carlo samples. It can be noticed that the Gamma distribution is very well represented, while the uniform distribution approximation has overshoots at the end of its support. These can be thought of as a Gibbs phenomenon in stochastic space and have also been observed by other authors [7].

5. Numerical experiments

Cunningham et al. [22] did experiments for biofilm growth and determined a set of values for the kinetic parameters for the bacterium *Pseudomonas aeruginosa*. In their paper no endogenous decay rate is used and the resulting Monod kinetics coefficients are $\mu_{\mathrm{m}} = 0.05$, $K_s = 0.32$, and $\mu_y = -0.3125$; the initial values of the concentrations are $c_M(t = 0) = 0.043$ and $c_N(t = 0) = 0.4$. We investigate the effect of randomness in the coefficients by considering a set of four cases:

- *Case* 1. All three parameters have uniform distributions: $K_s \sim U(0.28, 0.36)$, $\mu_{\max} \sim U(0.04, 0.06)$, $\mu_y \sim U(-0.325, -0.3)$. Legendre chaos expansion used.
- *Case* 2. $K_s \sim U(0.28, 0.36)$ and the other two parameters have Gaussian distributions with the same means and variances as in case 1. Hermite chaos expansion used.
- *Case* 3. $K_s \sim \mathrm{Gamma}(3.2, 0.1)$, leading to the same mean as in the above cases but a larger variance of $\mathrm{Var}[K_s] = 0.032$. The other two parameters as in case 2. Hermite chaos expansion used.
- *Case* 4. $K_s \sim \mathrm{Gamma}(3.2, 0.1)$ as in case 3; the other two parameters have uniform distributions as in case 1. Hermite chaos expansion used.
- *Case* 5. K_s has the Beta distribution with parameters $\alpha = 2$, $\beta = 4.25$, leading to the same mean but variance $\mathrm{Var}[K_s] = 0.03$. The other parameters as in case 1. Hermite chaos expansion used, with the coefficients of the expansion for the Beta distribution computed numerically.

In all five cases the chaos dimension, as well as the order of the polynomial chaos expansion, is equal to three.

Fig. 2 is a plot of the concentrations of bacteria and nutrients obtained from solving Monod's model using the Runge–Kutta–Fehlberg scheme; the coefficients are taken to be deterministic and set to their mean values. Obviously, this case corresponds to the case of random coefficients with zero variance. Fig. 3 shows the results obtained by Legendre chaos for the case of uniform distributions. In this and the next three figures we also plot the standard deviation interval, i.e. plot

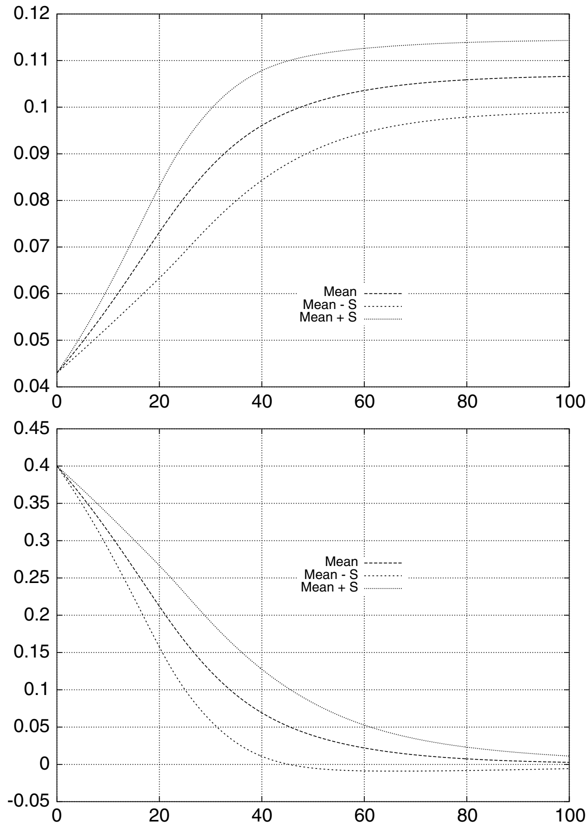

Fig. 6. Results for the mean concentrations and their variance for Case 4. Top figure: c_M. Bottom figure: c_N.

the curves $c_M - \sqrt{\mathrm{Var}[c_M]}$ and similarly for c_N. The value of the variance can be computed from the coefficients in the expansion. For c_M for example, the variance is

$$\mathrm{Var}[c_M(t)] = \sum_{i=1}^{P} c_{Mi}^2(t) * \mathrm{Var}[\Phi_i]. \tag{19}$$

Fig. 4 shows the results obtained for Case 2. There is very little quantitative difference between these results and the results for Case 1. In view of this, we find it reasonable to use a normal distribution as an approximation to a uniform distribution (and consequently use the Hermite chaos expansion) when the decision regarding the choice of the basis functions is less obvious.

The introduction of the Gamma distribution and the larger value of the variance causes the results for Case 3, shown in Fig. 5, to be considerably different from the first two cases. The largest impact is on the distribution of the bacterial concentration c_M, but the distribution of the nutrient concentration is however affected as well. Note that the standard deviation interval encompasses a range of negative values for the latter in this case at the end of the time interval. This may be considered an artifact of the straightforward numerical treatment which does not enforce the non-negativity of the amount of nutrients. Results for the last two cases are similar to those obtained for the third case, to the extent that hardly any difference can be noticed in the plots. This indicates, at least in the cases we studied, that knowing the values of the mean and variance of the parameters is of major importance; once these are set, the effect of the actual distribution seems to have a lower impact. Figs. 6 and 7 show the results for Cases 4 and 5.

As a final remark let us mention that, although the integration was performed for a sufficiently long time to reach the growth limiting plateau, no stability problems have been noticed. This is known to be a problem in the use of polynomial chaos, as the nonlinearity in the equations increases the high-frequency content of the solution with time, which is thus not captured accurately with a constant-order expansion. Nevertheless, we plan to investigate the impact of using different chaos orders (i.e. the convergence of the expansion) in future work.

6. Conclusions

There are many available models for microbial growth. The Monod model is a relatively simple model that still incorporates the effect of the amount of available nutrients into the growth rate, thus limiting the growth in a natural

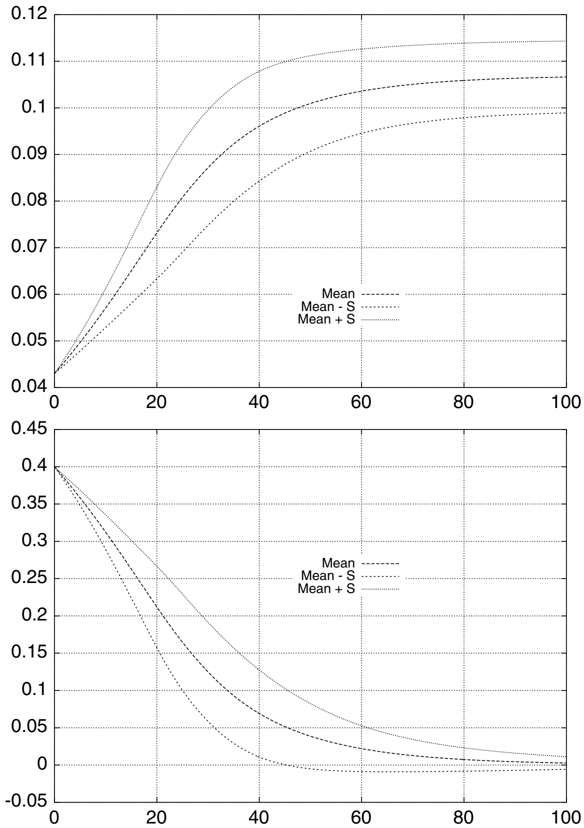

Fig. 7. Results for the mean concentrations and their variance for Case 5. Top figure: c_M. Bottom figure: c_N.

way. It is thus more realistic than the exponential or logistic growth model, hence also more complicated. Real data entering such a growth model is hard to obtain and may be expected to have large variability. The scattering of data is due to changes in population sizes, experimental conditions, errors and other uncertainties. Incorporating randomness in a population growth model is a realistic alternative to deterministic models which accounts for this variability. In this paper, we have modified Monod's model to take into account such randomness in the coefficients. The resulting system of random coefficient differential equations in time was solved approximately using the method of polynomial chaos. Several different distribution functions have been used for the random coefficients. Time integration over the periods of time of interest did not expose any evidence of instability. This makes the method a good candidate for the more complicated study of biofilm growth and convection in an aquifer that we plan to undertake in further work.

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